Anaconda and Jupyter Installation Tutorial for Windows

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Introduction

This tutorial will guide you through installing Anaconda, setting up a new Conda environment, installing Jupyter, and running your first Jupyter Notebook on Windows.

1 Step 1: Downloading the Anaconda Installer

1. Visit the official Anaconda website: `https://www.anaconda.com/products/individual`.
2. Click on **Download for Windows (Python 3.x, 64-bit)**.
3. Once the download is complete, run the `.exe` installer file.

2 Step 2: Installing Anaconda

1. **Select Installation Type**: Choose "Just Me" to install Anaconda only for your user account.
2. **Choose Installation Location**: Leave the default location unless you have a specific preference. Avoid installing in a directory path that contains spaces or special characters.
3. **Advanced Options**:
   - Uncheck the option to add Anaconda to the system's PATH environment variable (recommended). This avoids potential conflicts.
   - Check the box to register Anaconda as your default Python version.
4. Click **Install** and wait for the installation to complete.

3 Step 3: Verifying the Installation

1. Open the **Anaconda Prompt** (search for it in the Start Menu).
2. Verify the installation by typing the following command: `conda --version`.

4 Step 4: Creating a New Conda Environment Based on the Base Environment

Conda allows you to create isolated environments for different projects. We will create a new environment that clones the `base` environment to ensure you have the essential packages pre-installed.

4.1 Create and Activate the Environment

1. Open the **Anaconda Prompt**.
2. Create the environment by typing: `conda create --name myenv --clone base`.
3. Activate the environment by typing: `conda activate myenv`.

4.2 Explanation of the Conda Environment Setup

- `--name myenv`: Specifies the name of your new environment. In this case, it is called `myenv`, but you can choose any name.
- `--clone base`: Clones the existing `base` environment, ensuring your new environment starts with the same minimal packages.

4.3 Installing Jupyter

- Load your custom environment (if you have not done so previously): `conda activate myenv`.
- Install Jupyter by typing: `conda install jupyter`.

4.4 What is Jupyter?

Jupyter Notebook is an open-source web application that allows you to create and share documents containing live code, equations, visualizations, and narrative text. It is commonly used in data science, machine learning, and scientific computing to run Python code in an interactive, readable format.

The key features of Jupyter include:

- Executing Python code in real-time and seeing immediate outputs, such as graphs, tables, or text.
- Sharing the notebook with others so they can reproduce your analysis.
- Combining code with rich text, enabling you to explain your work and share insights interactively.

Jupyter Notebooks are widely used in education, research, and industry due to their interactive and user-friendly nature.

5 Step 5: Opening and Running Jupyter Notebook

5.1 Starting Jupyter Notebook

1. In the **Anaconda Prompt**, make sure the environment is active: `conda activate myenv`.
2. Start Jupyter Notebook by typing: `jupyter notebook`.

5.2 Creating a New Notebook

1. In the Jupyter interface, click on **New** (upper-right corner).
2. Choose **Python 3** under the "Notebook" section.

5.3 Running Python Code in Jupyter Notebook

1. In the new notebook, you will see an empty code cell that looks like this: `In [ ]:`.
2. Click inside the cell and type `print('Hello, World!')` and press `Shift + Enter` to run it.

5.4 Saving Your Work

To save your notebook, click **File** → **Save Notebook** or press `Ctrl + S`.

6 Step 6: Closing the Jupyter Notebook

1. Save your work if you have not already.
2. To close the notebook, click **File** → **Shutdown**.
3. Stop the Jupyter server by closing the terminal or pressing `Ctrl + C`.